



Impact of Super Resolution on the Detection of Plastic Bottles in Nature

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Environmental Challenge: Plastic Pollution Detection

 Plastic bottles are among the most common waste in nature

 Real-world detection systems use low-quality images (drones, distant cameras)

 Challenge: Small, blurry objects are hard to detect accurately

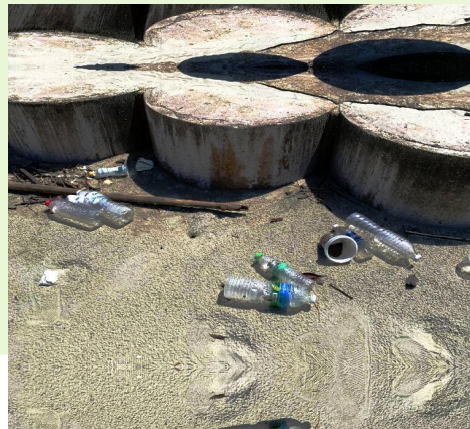


First Dataset Challenges



Problems

- Very small objects (few pixels only)
- Repetitive images from same location
- Minimal variation in lighting/angles



Solution

- Switched to Roboflow dataset
- More diverse scenes and conditions
- Better suited for both detection and SR

Methodology

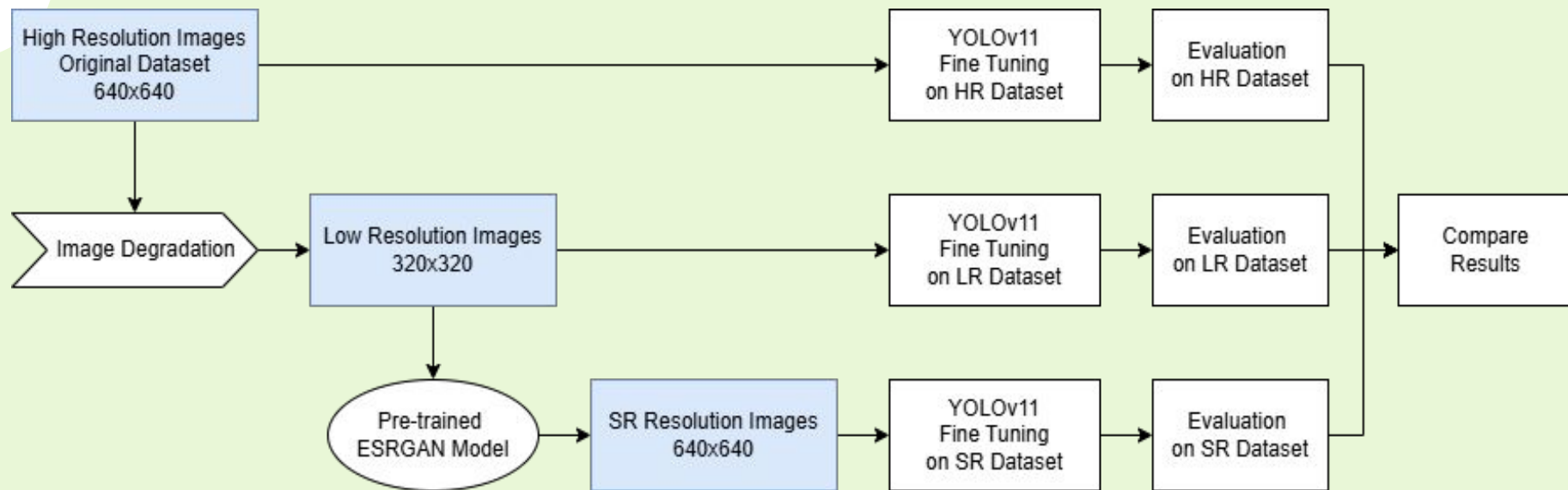


Image Degradation Process



High Resolution Image



Low Resolution Image

Downscaling

Bicubic interpolation
(640×640 → 320×320)

Blurring

Gaussian blur (5×5 kernel,
 $\sigma=1.0$)

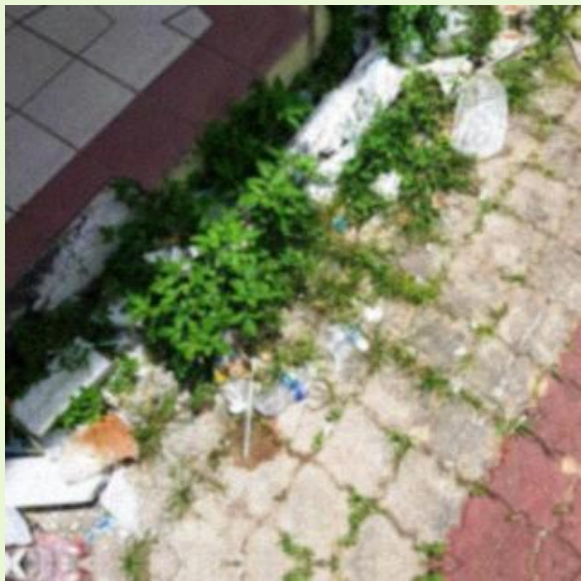
Noise

Gaussian noise ($\sigma=5$)

Super Resolution Enhancement

- Custom ESRGAN → 8 hours training, mediocre results
- Pre Trained Real-ESRGAN → 20 minutes, superior quality

Low Resolution



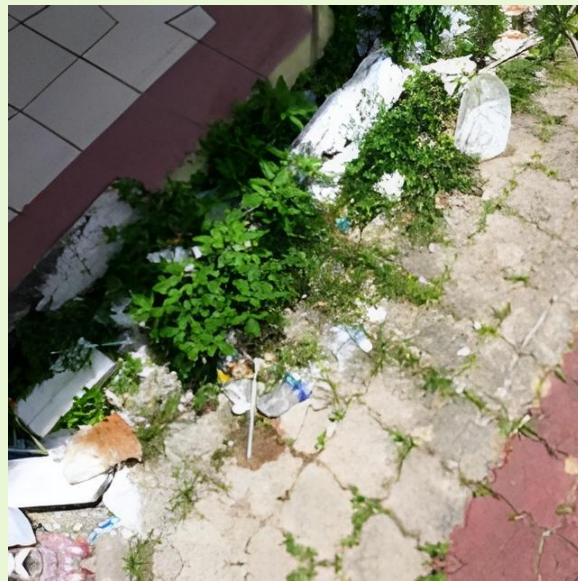
320x320

Custom SR-Model



640x640

Pretrained SR-Model



640x640

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Pretrained SR-Model





Training Configuration

01

Model Details

- YOLOv11 for all experiments
- 50 epochs training
- Adam optimizer ($lr=0.001$)
- Batch size: 16

02

Input Resolutions

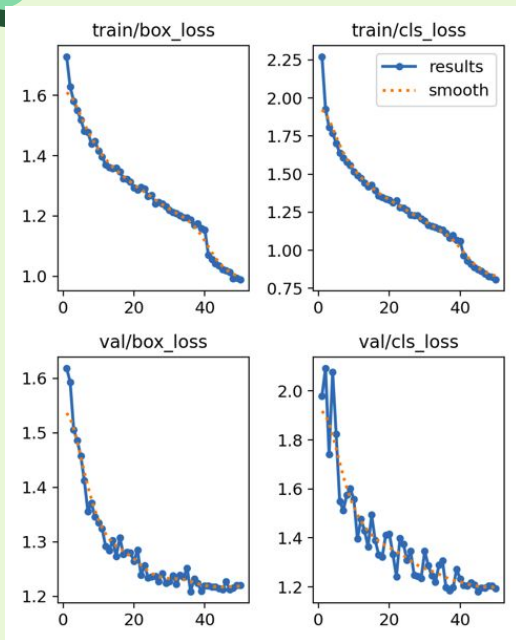
- HR & SR: 640×640
- LR: 320×320
- Fair comparison approach

03

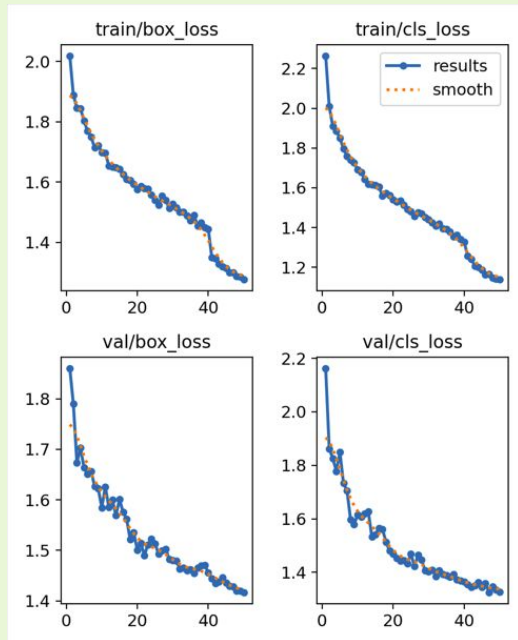
Evaluation Metrics

- mAP@0.50
- mAP@0.50:0.95
- Precision & Recall
- SSIM for image quality

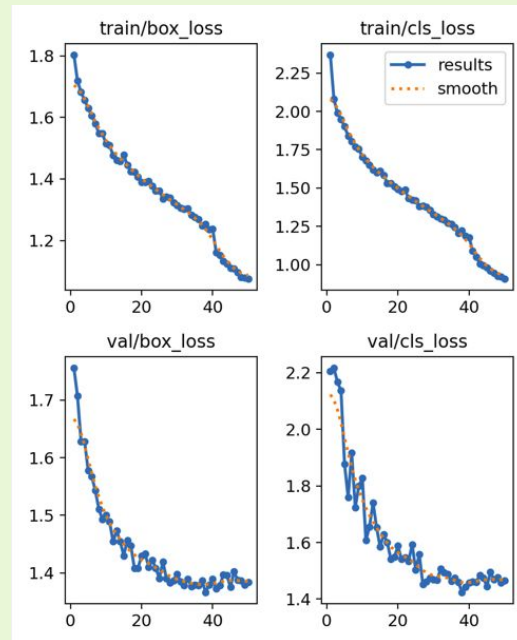
Train and Validation Losses



HR Dataset



LR Dataset



SR Dataset



Performance Comparison Results

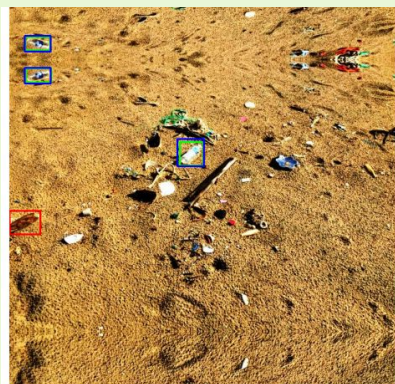


- SR achieves +5.4% mAP@50 improvement over LR
- SR precision nearly matches HR performance (0.744 vs 0.748)
- SSIM improved from 0.3605 to 0.4436

Dataset	Precision	Recall	mAP@50	mAP@50-95
High Resolution	0.748	0.546	0.671	0.499
Low Resolution	0.561	0.418	0.518	0.373
Super Resolution	0.744	0.411	0.572	0.418



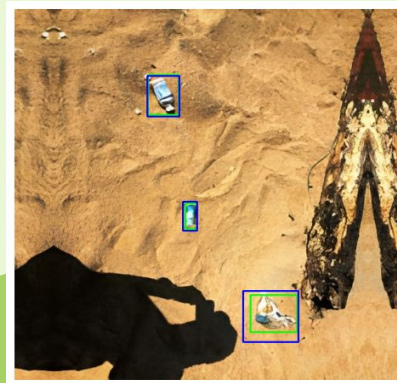
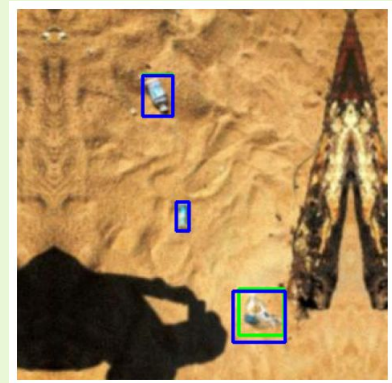
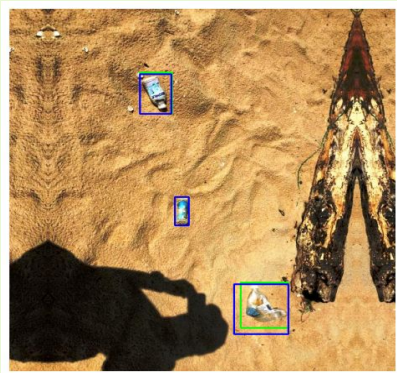
SR misses some detections that LR catches



SR improves true positive rate



SR reduces false positives



SR helps recover missed detections



Conclusions & Future Work



Key Achievements

- SR improved detection over LR
- Pretrained models more efficient than custom training
- Practical pipeline for real-world applications

Limitations

- SR still below HR performance
- Domain-specific training could help
- Some false negatives remain

Future Directions

- Train ESRGAN specifically on plastic waste images
 - Test other SR methods
 - Larger and more diverse datasets
 - Multi-scale training approaches
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